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**DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING AY
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SERVICE LEARNING PROJECT SYNOPSIS

ON

**“Web-Based Platform for Faculty Work Management and
AI-Assisted Student Book Review”**

**BACHELOR OF TECHNOLOGY COMPUTER SCIENCE &
ENGINEERING**

Submitted By

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INTRODUCTION

Schools and educational organizations require simple digital systems to manage routine faculty work, review submitted activities, and provide timely feedback. At the same time, reading activities often involve handwritten summaries that are time-consuming to review manually, especially when many students submit work.

This project proposes a web-based platform with separate interfaces for administrative and faculty users. The administrative interface will support the submission, review, approval, and feedback of daily faculty work. The faculty interface will support the review of handwritten student book summaries with assistance from a locally deployed AI system.

For book-summary evaluation, the system will extract text from student handwriting and evaluate the response against a teacher-defined standard for the selected book. The standard can contain flexible criteria and mark allocations defined by the faculty member, allowing the same evaluation framework to be applied consistently across multiple student submissions.

PROBLEM STATEMENT

- **Manual Faculty Work Review:** Daily faculty activities may be recorded and reviewed through fragmented or manual processes, making it difficult to maintain a consistent record of approvals and feedback.
- **Time-Consuming Handwritten Review:** Reviewing handwritten book summaries for a large number of students requires considerable faculty time and effort.
- **Inconsistent Evaluation:** Without a clearly defined and reusable standard, evaluation of student summaries may vary between submissions.
- **Limited Flexibility in Assessment:** Different books and learning objectives may require different evaluation criteria, which should be configurable by faculty rather than fixed in the system.
- **Resource Constraints for AI Processing:** The AI component should be capable of operating locally with limited computational resources and should not depend on a continuously available external AI service.

These issues increase the administrative workload, reduce the efficiency of handwritten assessment, and make it harder to maintain a structured record of faculty work and student reading activities.

OBJECTIVE

The main goal is to develop a lightweight web-based platform that streamlines faculty work review and assists educators in evaluating handwritten student book summaries using locally deployed AI.

The specific objectives are:

- **Administrative Review:** Provide an interface for authorized administrators to view, approve, reject, and provide feedback on daily faculty work submissions.
- **Faculty Book Review:** Provide an interface for faculty to manage student book-summary submissions and review AI-assisted evaluation results.
- **Handwriting-to-Text Processing:** Extract the content of handwritten student summaries using an appropriate local document or handwriting recognition component.
- **Flexible Evaluation Standards:** Allow faculty to define a reusable standard for a book, including criteria, descriptions, and mark allocations as required.
- **AI-Assisted Grading:** Compare extracted student content against the teacher-defined standard and generate a structured evaluation and suggested score.
- **Faculty Verification:** Allow faculty to review AI-generated results and, where necessary, modify or confirm the final assessment.
- **Modular Design:** Build the system so that the AI/OCR components, evaluation method, and user interfaces can be modified or replaced without redesigning the complete platform.

REQUIREMENTS

1. Authentication & User Roles

- Secure authentication with role-based access for administrators and faculty.
- Administrators can access faculty work records and review functions.
- Faculty can access student book-review and assessment functions.

2. Faculty Work Management

- Faculty can submit daily work/activity records.
- Administrators can view pending submissions and their details.
- Administrators can approve submissions, request changes, and provide feedback.
- Maintain a history of submissions, decisions, and feedback.

3. Student Book Summary Review

- Faculty can create or select a book and receive student summary submissions.
- Support image-based submission of handwritten student summaries.
- Process handwritten content through a locally deployed text/document extraction component.
- Store extracted text and associated submission information for review.

4. Flexible Grading Standard

- Faculty can define a grading standard for a selected book or reading activity.
- Standards can contain custom criteria, descriptions, weights or marks, and a total score.

- A defined standard can be reused for multiple student submissions.
- Standards can be preserved or versioned so previous evaluations remain associated with the standard used.

5. AI-Assisted Evaluation

- Use a locally deployed AI/LLM component to evaluate extracted student content against the teacher-defined standard.
- Generate structured criterion-wise scores, supporting observations, and feedback where appropriate.
- Keep the evaluation process configurable so the underlying AI model or processing component can be changed later.
- Provide the faculty member with the opportunity to verify, accept, or modify the suggested assessment.

6. Review & Records

- Display submitted work, extracted content, grading criteria, AI assessment, and final faculty decision.
- Maintain records of student submissions and their corresponding evaluations.
- Provide basic filtering or history views by book, student, faculty member, status, or date as required.

7. UI/UX Design

- Web-based responsive interfaces for administrative and faculty users.
- Clear workflows for submission, review, approval, and assessment.
- Simple interfaces that minimize unnecessary steps for faculty and administrators.
- The design should remain modular so that additional user interfaces or access methods can be introduced later.

TECH STACK

Layer	Technology / Approach
Frontend	React.js with Vite
Styling	Tailwind CSS
Backend	Node.js with Express.js
API	RESTful API
Authentication	JWT-based authentication with role-based access control
Database	PostgreSQL
Deployment	Vercel
AI / OCR	Local AI pipeline for handwriting/text extraction and rubric-based evaluation, using a lightweight LLM and suitable local inference runtime (e.g., Ollama or llama.cpp)

SUMMARY

This project aims to provide a centralized platform for managing faculty work and assisting educators in reviewing student book summaries. The system separates administrative review from faculty-oriented student assessment while maintaining a common backend for authentication, records, and processing.

A key component of the platform is the AI-assisted evaluation of handwritten student summaries. Instead of using a fixed grading rule, faculty members can define a flexible standard for a book or reading activity. The system can then extract the student's written content, evaluate it against the selected standard, and present a structured suggested assessment for faculty verification.

The proposed design intentionally keeps the AI model, user interface, and deployment approach modular. This allows the implementation to adapt to available hardware, future requirements, and alternative AI or interaction methods while maintaining the core workflow of faculty review and student assessment.